

Is the Grass Greener Next Door? Evidence from Walking Accessibility in Barcelona

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Abstract

Can all of Barcelona’s residents walk to a doctor, a school, and a park within 15 minutes—or does that depend on how much they earn? We geocode 28,983 service points across the city’s 73 neighbourhoods and estimate a Spatial Durbin Model to answer this question. We find that a 10% increase in neighbourhood income predicts about 8% more services within walking distance, but this gap comes entirely from healthcare, schools, and parks—not from groceries or public transit. Roughly half of the total income effect spills over from neighbouring areas, so a neighbourhood’s service access depends on who lives next door, not just on its own characteristics. When we measure services per capita instead of in absolute counts, the income gradient vanishes: wealthier areas attract more services *and* more people. Barcelona’s 15-minute city agenda should target specific shortfalls—healthcare and schools especially—in the peripheral, low-income clusters we identify.

Contents

1	Introduction	3
2	Conceptual Framework	4
2.1	Spatial Economics and the Gravity Equation	4
2.2	Sorting: Why Income and Accessibility Go Together	4
2.3	Related Literature	5
3	Data	6
4	Empirical Strategy	9
5	Results	10
5.1	Spatial Autocorrelation	10
5.2	Model Comparison	11
5.3	SDM Estimates and Impact Decomposition	12
6	Extensions and Robustness	13
6.1	Gravity-weighted Accessibility	13
6.2	Market Access as a Structural Control	14
6.3	Accessibility inequality: Gini and Lorenz curve	14
6.4	Per-capita Service Adequacy	15
6.5	Which Services Drive the Gap?	16
6.6	Non-linearity and Alternative Threshold	17
7	Limitations	18
8	Conclusion	19
9	References	20

1 Introduction

Imagine two residents of Barcelona: one in Sarrià–Sant Gervasi, one of the city’s wealthiest districts, and another in Trinitat Vella, among its poorest. Both hear the promise of a walkable city under Barcelona’s *superilla* programme. But when we count the essential services within a 15-minute walk of each, we find a large gap. Our Spatial Durbin Model puts a number on it: a 1% increase in per-capita household income predicts 0.38% more services nearby ($p = 0.048$). Once we factor in spatial spillovers—how neighbours’ income also shapes your own accessibility—the total income elasticity rises to 0.80 ($p = 0.002$). Accessibility clusters strongly in space (Moran’s $I = 0.43$, $p = 3.3e-10$), and formal model-selection tests overwhelmingly favour specifications that include neighbours’ characteristics ($p = 6.9e-06$).

The 15-minute city concept, popularised by Moreno et al. (2021), now guides compact-city planning worldwide. Barcelona’s *superilla* programme operationalises it by redesigning streets and public spaces so that residents can walk to groceries, healthcare, schools, parks, pharmacies, and transit within 15 minutes (Ajuntament de Barcelona, 2021). We ask a simple question: *does this actually work equally for everyone, or does income determine how walkable your neighbourhood is?*

We tackle this with 28,983 geolocated points of interest from OpenStreetMap, official income and population data from the city council, and a battery of spatial econometric models that we connect to the structural gravity equation and spatial equilibrium ideas from Redding (2016), Allen and Arkolakis (2014), and Donaldson (2018).

Three findings stand out. First, the income gap varies sharply across services—we see it strongly for health-care centres (elasticity 1.40), parks (0.75), and schools (0.66), but barely at all for groceries (0.11) and transit (0.18). Second, when we measure services per capita instead of in raw counts, income stops mattering ($p = 0.48$). Wealthier areas attract more services *and* more people. Third, a gravity-weighted accessibility measure inspired by trade theory tightens the fit: the spatial autoregressive parameter jumps from 0.32 to 0.50, and the R^2 improves to 0.75.

2 Conceptual Framework

2.1 Spatial Economics and the Gravity Equation

We ground our analysis in the quantitative spatial economics framework that Redding and Rossi-Hansberg (2017) survey. In these models, locations connect through trade frictions d_{ni} , and what happens in one place depends on what happens everywhere else. The price index in location n takes the form:

$$P_n = \gamma \left[\sum_{i \in S} A_i (w_i d_{ni})^{-\theta} \right]^{-1/\theta}$$

where A_i denotes productivity, w_i wages, and θ governs how quickly the influence of distant places dies off. A shock to one location propagates outward through the whole system. This is exactly what a spatial lag model captures in reduced form.

In our Spatial Durbin Model (SDM), the spatial multiplier $(\mathbf{I} - \rho \mathbf{W})^{-1}$ serves the same purpose: it channels the effects of local changes through the network of neighbouring areas. When we decompose each coefficient into direct and indirect effects, we separate a location’s own fundamentals from the spatial diffusion channel.

We take this a step further by building a gravity-weighted accessibility measure:

$$\text{GravAccess}_n = \sum_j e^{-\beta d_{nj}}$$

where j indexes all points of interest and β controls distance decay. Instead of drawing a hard line at 1,200 m and counting everything inside, we give more weight to nearby services and less to faraway ones—much like trade flows decline with distance in the structural gravity equation.

2.2 Sorting: Why Income and Accessibility Go Together

Why should income predict accessibility in the first place? The spatial equilibrium framework gives a clean answer: sorting. Workers pick where to live based on amenities, wages, and housing costs. In the canonical model (Redding, 2016), the probability that a worker chooses location n equals:

$$\lambda_n = \frac{B_n (v_n / P_n)^\epsilon}{\sum_{i \in S} B_i (v_i / P_i)^\epsilon}$$

If accessibility forms part of the amenity bundle B_n , then households who can afford higher rents sort into well-served neighbourhoods. We observe the equilibrium outcome of this process: rich people in areas with lots of services—not because income causes services to appear, but because both emerge jointly.

Diamond and Suarez Serrato (2025) find that high-education US workers increasingly sort into high-amenity areas rather than high-wage areas—amenities now dominate the pull. On the other side, Ellder (2024, 2025) shows that Swedish neighbourhoods which become 15-minute cities tend to gentrify, with housing prices rising by up to 30%. So sorting creates a feedback loop: better services attract richer residents, which pushes up prices, which pushes out poorer ones. Our cross-sectional model cannot capture this dynamic, but the SDM’s decomposition into own and spillover effects at least separates the two channels.

2.3 Related Literature

Abbiasov et al. (2024) used GPS data from 40 million US devices to quantify the 15-minute city and found that only 14% of consumption trips happen locally. They also uncovered a troubling trade-off: more local living correlates with more segregation for low-income residents.

Closer to our setting, Ferrer-Ortiz et al. (2022) mapped pedestrian accessibility in Barcelona using network analysis and showed that the compact core meets the 15-minute standard, but the periphery does not. Plaza-Herrera and Mercade-Aloy (2025) extended this to the metropolitan area and found that while 81% of the general population can reach all service categories, only 42% of the elderly can.

Thaury et al. (2024) studied Paris with a SARAR spatial model and OpenStreetMap data, finding the same centre-periphery story: lower-income families live in less-amenity-rich areas. Mouratidis (2024) wrote a pointed critique identifying seven pitfalls of the 15-minute city concept—among them, ignoring differences in walking ability across populations and conflating the *number* of services with their *adequacy*. We address both of these directly.

Finally, Caliendo et al. (2018) showed how productivity shocks propagate across US states through input-output linkages—a different application, but the same propagation logic that our SDM captures in the context of urban services.

3 Data

We combine three open data sources, all downloaded automatically in R.

Points of interest come from OpenStreetMap, queried through the `osmdata` package for six categories: groceries, pharmacies, schools, healthcare, parks, and transit stops (28,983 POIs in total). Neighbourhood boundaries (the 73 *barris*) and socioeconomic data come from the Ajuntament de Barcelona’s Open Data portal: disposable household income per capita for 2022 and population for 2025, both originally at the census-section level and aggregated up to the *barri*.

We construct the accessibility score by drawing a 1,200 m buffer around each neighbourhood centroid (roughly 15 minutes at walking speed) and counting all POIs inside it. The resulting score ranges from 92 to 2,367, with a mean of 1,628. The income index runs from 54 to 188 (Barcelona average = 100)—so the poorest neighbourhood earns about half the city average and the richest nearly double. We perform all spatial operations in EPSG:25831 (UTM zone 31N) for accurate metric distances.

Figure 1 maps the 28,983 POIs. A few patterns jump out immediately. Transit stops and parks blanket the city—Barcelona’s metro/bus network and green spaces cover most of the territory. Pharmacies and groceries spread fairly evenly, which makes sense: these market-driven services follow population. Healthcare facilities, however, cluster visibly in central and upper-city areas. This foreshadows what the regressions confirm: healthcare shows the sharpest income gradient.

Essential Services in Barcelona

Points of interest from OpenStreetMap

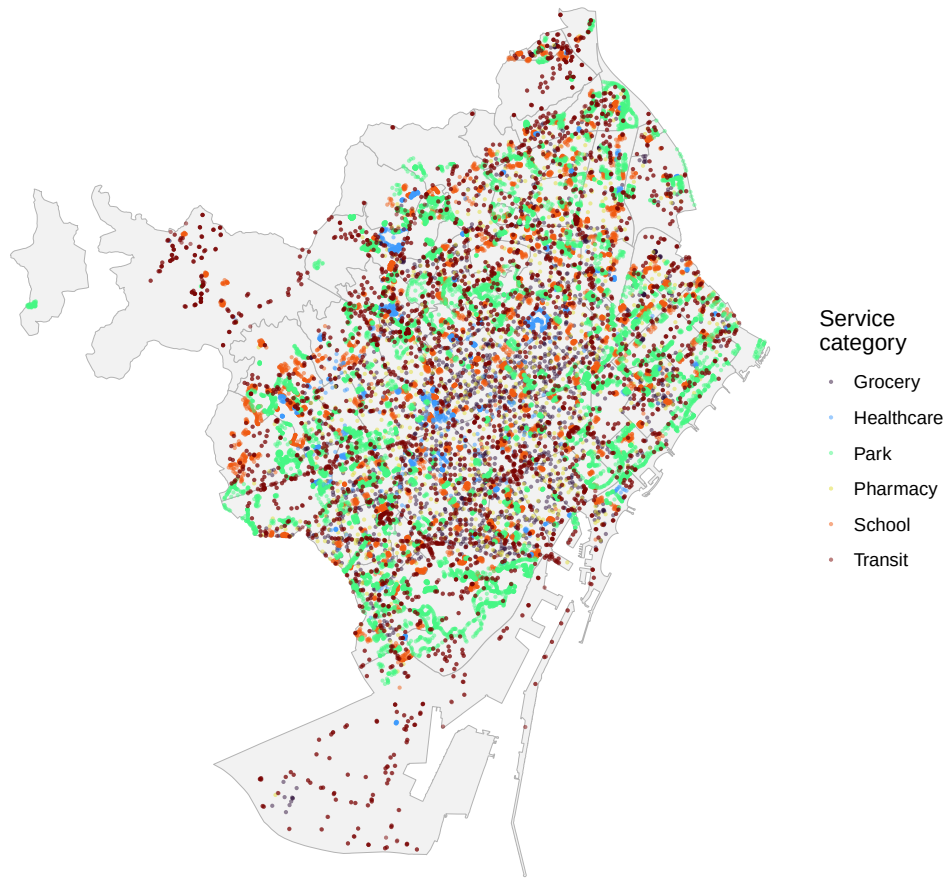


Figure 1: All 28,983 essential service POIs in Barcelona, coloured by category. Transit and parks blanket the city. Healthcare facilities (blue) cluster visibly in central and upper-city districts—a preview of the category-specific income gap we document later.

Figure 2 puts the accessibility and income maps side by side, and the overlap jumps out. The central Eixample district—Barcelona’s famous grid—stands out as both the richest and the most service-dense part of the city. The northern periphery (Nou Barris, parts of Sant Andreu) shows dark shading on both maps: low income, few services. This spatial co-movement motivates our analysis, though the maps alone cannot tell us how much income vs. population density vs. centrality drives it. That is what the regressions do.

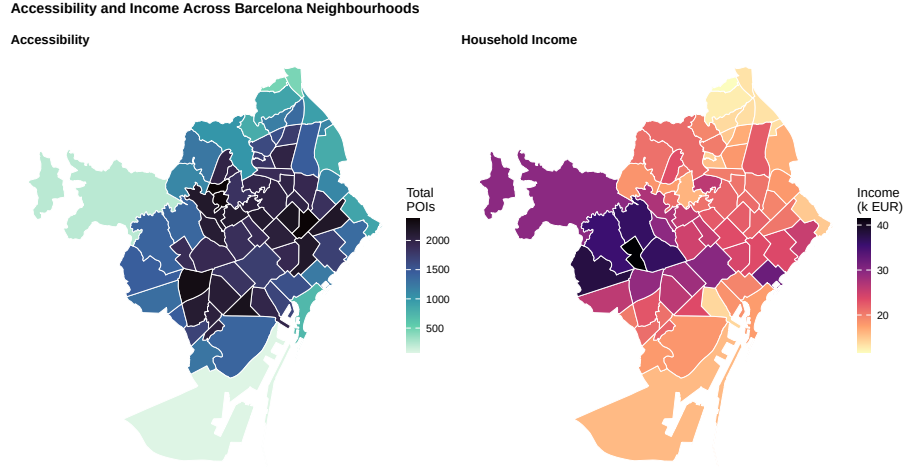


Figure 2: Accessibility (left) and household income (right) across Barcelona’s 73 neighbourhoods. Darker = lower. The visual overlap stands out: the low-income periphery is also where services thin out.

Figure 3 turns the two choropleths into a scatter plot. Each dot represents a neighbourhood (sized by population), and the OLS line slopes upward ($R^2 = 0.68$). The relationship does not hinge on outliers—the cloud tracks the line fairly consistently. Still, we spot some interesting exceptions: a cluster of middle-income Eixample neighbourhoods scores well above the trend line (they sit centrally and pack many services despite not being the richest), while a few wealthy but peripheral neighbourhoods fall below it.

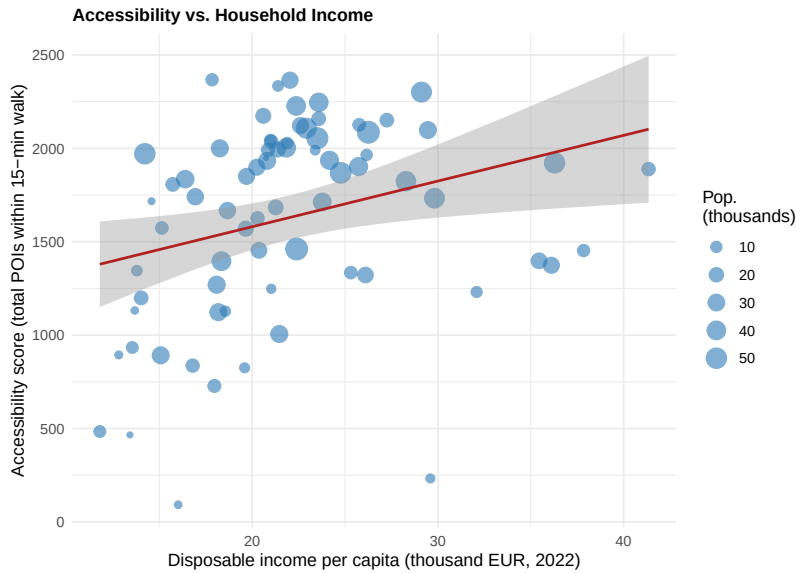


Figure 3: Accessibility vs. income. Each dot represents a neighbourhood (sized by population). The OLS line ($R^2 = 0.68$) confirms the positive relationship. Some dense, middle-income areas beat the trend thanks to their central location.

4 Empirical Strategy

We build spatial neighbours using queen contiguity (average: 5.2 neighbours per *barri*) and row-standardise the weights matrix so that the spatial lag becomes a simple average of neighbours' values. Our baseline log-log OLS takes the form:

$$\ln(\text{Access}_i) = \alpha + \beta_1 \ln(\text{Income}_i) + \beta_2 \ln(\text{PopDensity}_i) + \beta_3 \ln(\text{DistCenter}_i) + \varepsilon_i$$

Since both sides enter in logs, every coefficient is an *elasticity*: $\beta_1 = 0.55$ means a 1% increase in income associates with a 0.55% increase in accessible services.

We then estimate four spatial variants—SAR, SEM, SDM, SDEM—following LeSage and Pace (2009). The SDM serves as our workhorse:

$$\ln(\text{Access}_i) = \rho \sum_j w_{ij} \ln(\text{Access}_j) + \mathbf{x}'_i \beta + \sum_j w_{ij} \mathbf{x}'_j \theta + u_i$$

It includes both a spatial lag of the outcome ($\rho \mathbf{W}\mathbf{y}$) and spatial lags of all three covariates ($\mathbf{W}\mathbf{X}$). The SDM nests both SAR and SEM as special cases, so we are unlikely to misspecify. More importantly, it lets us split each coefficient into a *direct* effect (how a neighbourhood's own characteristics affect its own accessibility) and an *indirect* or *spillover* effect (how its characteristics spill over to neighbours). The spatial multiplier $(\mathbf{I} - \rho \mathbf{W})^{-1}$ drives this decomposition—the reduced-form version of the propagation channel in structural spatial economics (Caliendo et al., 2018).

We extend the analysis in two ways. First, we replace the buffer count with a gravity-weighted measure and re-estimate the SDM. Second, we swap the distance-to-centre control for a Harris (1954) market-access potential, $\text{MA}_n = \sum_{i \neq n} \text{Pop}_i / d_{ni}$, which captures a neighbourhood's position relative to the whole population distribution rather than its distance to one point.

5 Results

5.1 Spatial Autocorrelation

Global Moran's $I = 0.43$ ($p = 3.3e - 10$): accessibility clusters strongly in space. Neighbourhoods with lots of services sit next to other neighbourhoods with lots of services, and the same holds for underserved areas.

Figure 4 shows where these clusters form. The red areas (High-High) span a large contiguous block in the centre—Eixample, Gracia, parts of Sant Marti and Sants-Montjuic. These neighbourhoods have it all: high accessibility themselves and well-served neighbours. The blue areas (Low-Low) sit on the northern and western periphery—Nou Barris, Torre Baro, parts of Horta-Guinardo. Grey areas fail to reach statistical significance. What stands out is the near-absence of spatial outliers: we find almost no High-Low or Low-High neighbourhoods, which tells us the accessibility landscape forms a smooth gradient from centre to periphery, not a patchwork.

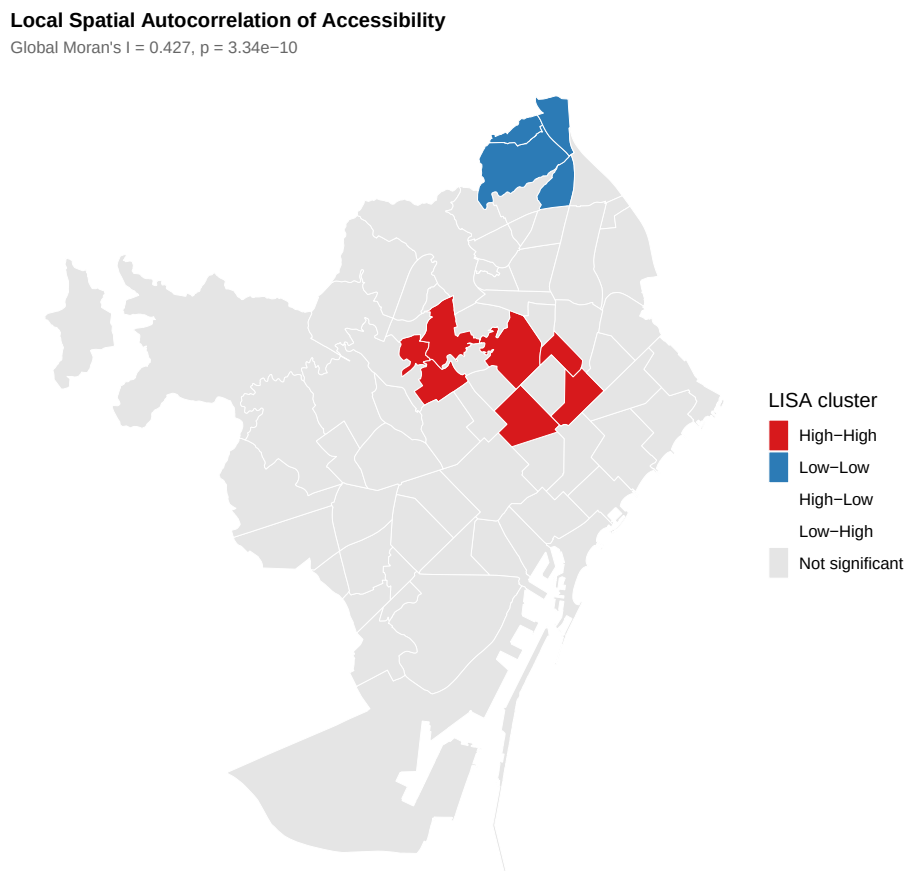


Figure 4: LISA cluster map. Red: High-High (well-served, well-served neighbours). Blue: Low-Low (underserved, underserved neighbours). Grey: not significant. The near-absence of spatial outliers points to a smooth centre-to-periphery gradient.

Figure 5 presents the same information as a scatter plot. Each neighbourhood's standardised accessibility score sits on the horizontal axis; the average of its neighbours' scores on the vertical. The positive slope

gives us Moran’s I . Most points cluster in the upper-right (High-High) and lower-left (Low-Low) quadrants, with few in the off-diagonal quadrants where outliers would appear.

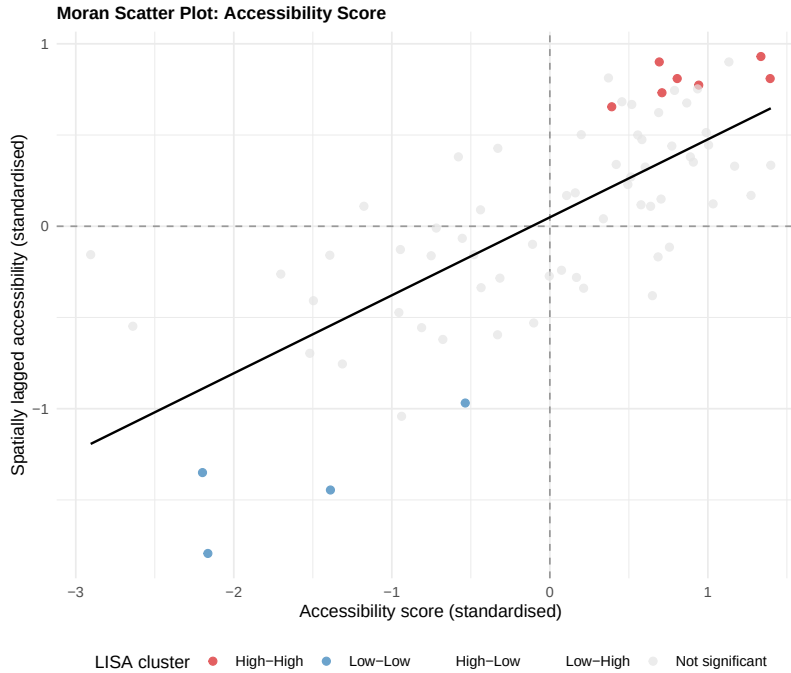


Figure 5: Moran scatter plot. Each neighbourhood plotted against its neighbours’ average. The positive slope ($I = 0.43$) and the clustering in the HH and LL quadrants confirm strong spatial autocorrelation.

5.2 Model Comparison

Table 1 compares the five specifications. The gap is dramatic. SDM and SDEM—the two models that include spatially lagged covariates ($\mathbf{WX}$)—score AIC values of 20.0 and 16.2, respectively. OLS, SAR, and SEM all hover around 40. The message: accounting for spatial dependence in the outcome or the errors alone does not suffice; what really matters is that *neighbours’ characteristics* (their income, their density, their centrality) predict a neighbourhood’s own service count.

Table 1: Model comparison. Lower AIC = better fit. The two models with spatially lagged covariates (SDM, SDEM) dominate by a wide margin.

Model	AIC	Log-Likelihood
SDEM	16.2	0.9
SDM	20.0	-1.0
OLS	39.7	-14.9
SAR	40.6	-14.3
SEM	40.7	-14.4

Likelihood ratio tests formalise this. The test statistic compares the log-likelihoods of a restricted model

(without **WX**) against an unrestricted one (with **WX**):

$$LR = 2 \times (\ell_{\text{unrestricted}} - \ell_{\text{restricted}})$$

For SDM vs. SAR:

$$LR = 2 \times (-1.0 - (-14.3)) = 26.7 \quad (p = 6.9e - 06)$$

For SDEM vs. SEM:

$$LR = 2 \times (0.9 - (-14.4)) = 30.5 \quad (p = 1.1e - 06)$$

Both follow a χ^2 distribution with 3 degrees of freedom (the three spatially lagged covariates added by the unrestricted model). We reject the restricted models decisively: the spatially lagged covariates matter. This echoes the prediction from spatial economics: in a world of interconnected locations, a neighbourhood’s fate depends on the fundamentals of the places it borders.

5.3 SDM Estimates and Impact Decomposition

We estimate $\hat{\rho} = 0.32$ ($p = 0.027$), meaning a moderate but significant share of a neighbourhood’s accessibility “comes from” its neighbours. Table 2 shows the impact decomposition.

Table 2: SDM impact decomposition (simulated SEs, $R = 500$). Direct = effect on own neighbourhood; Indirect = spillover from/to neighbours; Total = both combined.

Variable	Direct	Indirect	Total
ln(Income)	0.40**	0.40	0.80***
ln(PopDensity)	0.24***	0.20**	0.44***
ln(DistCenter)	-0.98***	1.26***	0.28**

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$

The total income effect equals 0.80. Because both variables enter in logs, this is an elasticity: a neighbourhood with 10% higher income has roughly 8% more services within walking distance. About half comes from the direct channel (0.40) and half from spillovers (0.40). The spillover part has a natural reading: wealthy households cluster together, and service providers respond to the aggregate demand of the cluster, not just of a single neighbourhood.

Population density matters even more (total effect 0.44). Services follow people—the more residents nearby, the more pharmacies, grocery stores, and bus stops appear. Both the direct and indirect channels reach significance, meaning that having dense neighbours also helps.

Distance to centre tells an interesting story. The direct effect is strongly negative (-0.98): sitting far from Plaça Catalunya hurts. But the indirect effect is strongly positive (1.26): if you live on the periphery, having central neighbours helps because their services sit nearby. These two effects largely cancel out (total 0.28). The SAR model, which cannot separate these channels, estimates a near-zero effect for distance—misleading. Our SDM reveals that distance matters a lot, but through two opposing channels.

6 Extensions and Robustness

6.1 Gravity-weighted Accessibility

Our baseline measure treats all services within 1,200 m equally and ignores everything beyond the cutoff. A pharmacy 50 m away counts the same as one at 1,199 m, and one at 1,201 m counts zero. The gravity-weighted alternative assigns each POI a weight that fades smoothly with distance: $\text{GravAccess}_n = \sum_j e^{-\beta d_{nj}}$ ($\beta = 0.001$). A service 700 m away gets about half the weight of one at zero distance.

Table 3 shows the SDM impacts with this measure. Spatial dependence strengthens substantially ($\hat{\rho} = 0.50$, $p < 0.001$). The income elasticity holds up (direct: 0.37, total: 0.66). The gravity measure captures something the buffer count misses: services just beyond 1,200 m still matter, and services very close by matter more than those at the buffer’s edge.

Table 3: SDM impacts with gravity-weighted accessibility. The spatial parameter ($\hat{\rho} = 0.50$) nearly doubles the baseline, and the income effect remains significant.

Variable	Direct	Indirect	Total
ln(Income)	0.37***	0.29	0.66***
ln(PopDensity)	0.21***	0.25***	0.47***
ln(DistCenter)	−0.69***	0.88***	0.19

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$

Figure 6 compares the gravity scatter to the baseline (Figure 3). The points hug the regression line more tightly ($R^2 = 0.75$ vs. 0.68\$), which tells us the gravity measure does a better job of capturing the income–accessibility relationship. This supports Mouratidis’s (2024) argument that sharp distance thresholds introduce unnecessary noise.

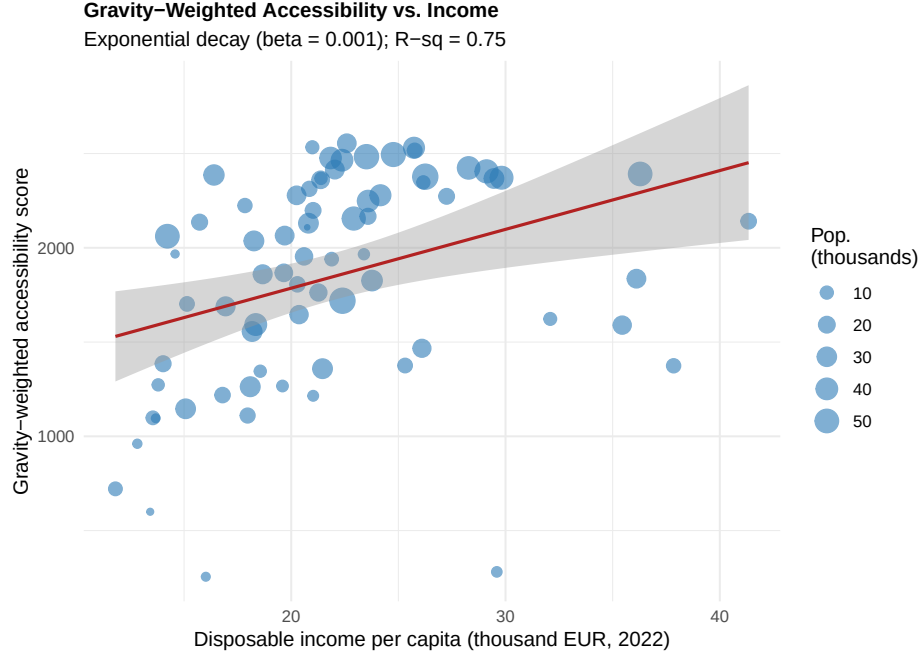


Figure 6: Gravity-weighted accessibility vs. income. The tighter fit ($R^2 = 0.75$) compared with Figure 3 ($R^2 = 0.68$) shows the continuous distance kernel captures the income gradient better than the sharp 1,200 m cutoff.

6.2 Market Access as a Structural Control

Our baseline uses distance to Plaça Catalunya as the centrality control, but why should one point matter? A neighbourhood’s position relative to the *entire population* drives service provision. We replace log distance-to-centre with the Harris (1954) market-access potential, $MA_n = \sum_{i \neq n} \text{Pop}_i / d_{ni}$. The OLS fit improves from $R^2 = 0.68$ to $R^2 = 0.79$, and market access enters with high significance ($p = 1.7e - 07$). The income gradient survives at 0.31 ($p = 0.009$): the gap is not just a centrality artefact.

6.3 Accessibility inequality: Gini and Lorenz curve

How unequal is the spatial distribution of services? The Gini coefficient for total accessibility equals 0.177—moderate inequality, not extreme. But the per-capita Gini jumps to 0.521, reflecting the wide range of population densities.

The category breakdown reveals a clear pattern. Healthcare has the highest Gini (0.459)—the most unequally distributed service. Grocery (0.390) and pharmacy (0.295) show moderate inequality. Transit (0.175) and parks (0.207) distribute most evenly, thanks to the metro/bus network and Barcelona’s scattered green spaces.

Figure 7 shows the population-weighted Lorenz curve. If every neighbourhood had services in proportion to its population, the curve would sit on the 45-degree line. Instead, it bows below: the bottom 40% of the population (ranked by accessibility) has access to roughly 25% of services. The shaded area gives the visual counterpart of the Gini coefficient.

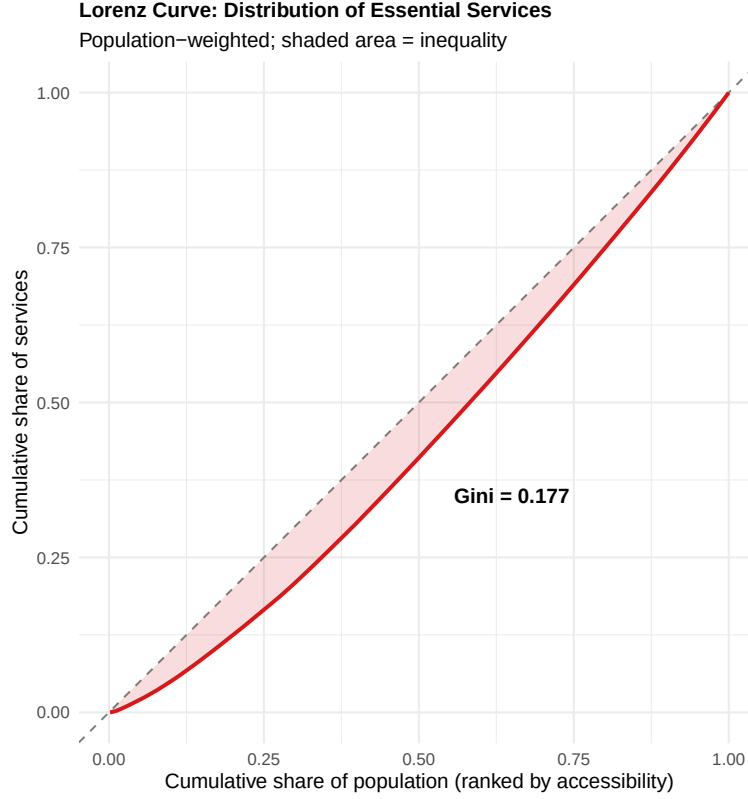


Figure 7: Population-weighted Lorenz curve. The 45-degree line = perfect equality. The curve bows below it: residents of the least accessible neighbourhoods get a disproportionately small share of the city’s services.

6.4 Per-capita Service Adequacy

A neighbourhood with 2,000 service points sounds great—until you learn it has 60,000 residents. Mouratidis (2024) and UBE (2024) warn against confusing the *number* of services with their *adequacy*. When we run the OLS with POIs per 1,000 residents as the outcome, the income coefficient flips to -0.25 and loses significance ($p = 0.48$; $R^2 = 0.12$). Wealthier areas have more services but also more residents. Per capita, the playing field is roughly level.

Figure 8 makes this vivid. Compare it to the left panel of Figure 2: the spatial pattern differs. Some peripheral neighbourhoods that looked underserved in absolute terms now show perfectly reasonable per-capita scores, because their low population means fewer people share the available services. Meanwhile, some of the densest central areas, which scored highest in absolute terms, drop to middling levels per capita.

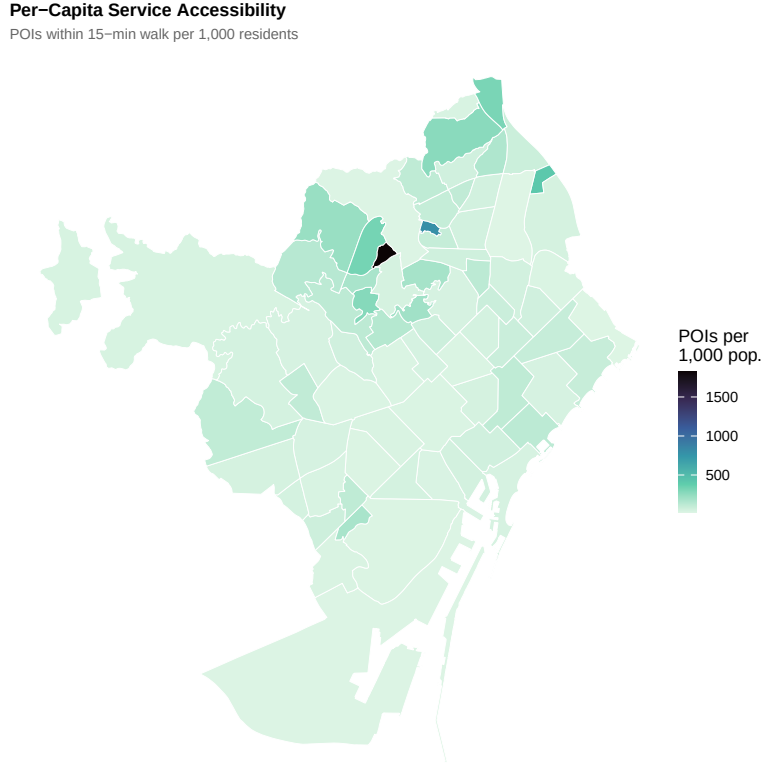


Figure 8: Per-capita accessibility (POIs per 1,000 residents). Compare with the left panel of Figure 2: the story changes. Some peripheral areas turn out well served relative to their population; some dense central areas do not.

6.5 Which Services Drive the Gap?

Table 4 runs a separate OLS for each service category. The numbers tell a clear story.

Table 4: Income elasticity by service category (OLS). An elasticity of 1.40 for healthcare means that a 10% income increase predicts a 14% increase in healthcare facilities within walking distance.

Category	Elasticity	Std. Error	<i>p</i> -Value
Healthcare	1.40	0.42	0.001
Park	0.75	0.34	0.030
School	0.66	0.23	0.006
Pharmacy	0.32	0.21	0.133
Transit	0.18	0.12	0.137
Grocery	0.11	0.23	0.643

Healthcare has the largest elasticity (1.40, $p = 0.001$). In plain language: a neighbourhood with 10% higher income has about 14% more clinics and doctors nearby. Parks and schools also show significant income gradients. All three are predominantly *public* services, whose locations municipal planners decide—not market demand alone.

Groceries (0.11) and transit (0.18) show no significant income link. Grocery stores go wherever people buy

food, regardless of neighbourhood income. Transit follows network logic: Barcelona’s metro and bus lines aim for coverage, not for serving wealthy areas.

Figure 9 plots the point estimates with 95% confidence intervals. Red dots (significant) sit to the right of zero; grey dots (not significant) straddle it. The visual gap between the top three categories and the bottom three is unmistakable.

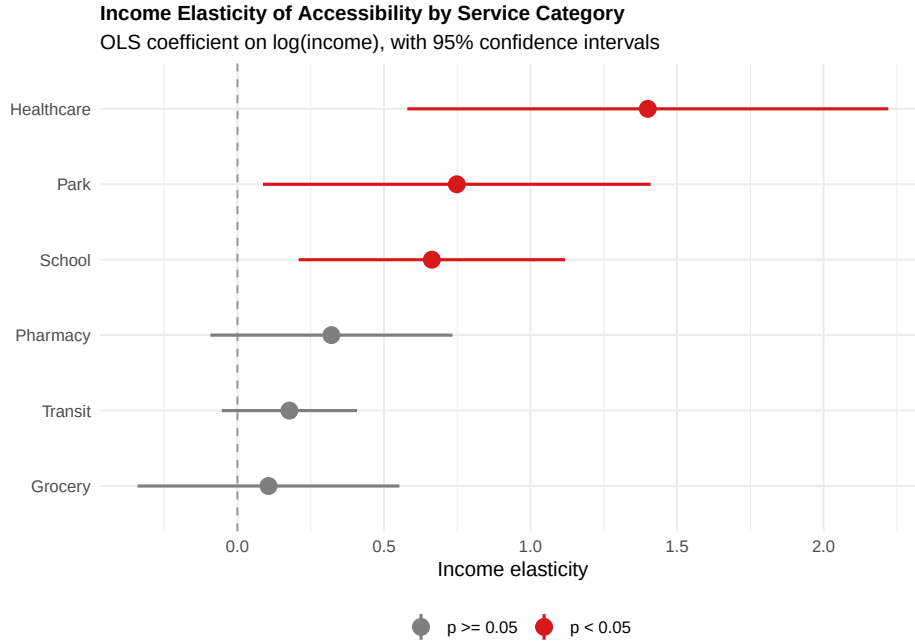


Figure 9: Income elasticity by service category with 95% CIs. Red = significant at 5%. The gap between publicly-provided services (healthcare, parks, schools) and market/network-driven services (grocery, transit) stands out.

6.6 Non-linearity and Alternative Threshold

We add a quadratic term in log income and find a negative coefficient (-0.66 , $p = 0.083$), suggesting some concavity: the relationship steepens at the bottom. This encourages policy action—investments in the poorest areas would generate the largest accessibility gains.

To verify that our results do not hinge on the 1,200 m threshold, we re-run everything with an 800 m buffer (about a 10-minute walk). The OLS income elasticity comes in at 0.48 ($p = 0.001$) and the SDM gives $\hat{\rho} = 0.27$ ($p = 0.069$). The conclusions hold.

7 Limitations

We want to be upfront about six boundaries of this analysis.

First, we cannot make causal claims. We document a robust income–accessibility association, but sorting works both ways: wealthier residents may choose well-served neighbourhoods, and well-served neighbourhoods may attract wealthier residents. We lack an instrument to break this simultaneity, a limitation we share with Thaury et al. (2024) and most of the 15-minute city literature.

Second, omitted variables could bias our estimates. We control for population density and centrality, but other unobserved factors—housing age, zoning regulations, tourism intensity—could drive both income and accessibility. Tourist-heavy areas like the Gothic Quarter have many services because of visitors, not because residents earn more, and we cannot separate tourist-serving from resident-serving POIs.

Third, our accessibility measure is crude. We use Euclidean buffers rather than street-network distances, count all POIs equally regardless of capacity, and rely on crowdsourced OpenStreetMap data that may have uneven coverage across neighbourhoods.

Fourth, aggregation matters. Working at the *barri* level means we use centroids to represent entire neighbourhoods that vary enormously in size (1–9 km²) and population (1,000–60,000). Finer-grained census-section data would help, but public income data does not go below the *barri*.

Fifth, we ignore dynamics. Our cross-section captures a single moment. We cannot track whether the income gradient is widening or narrowing, nor whether gentrification—which Ellder (2024) documents in Swedish 15-minute cities—is reshaping Barcelona’s accessibility map.

Sixth, the sample is small. Seventy-three observations limit statistical power. The SDM estimates eight parameters, leaving about 65 degrees of freedom, so borderline results (like the quadratic income term at $p = 0.083$) deserve caution.

None of these limitations overturns our central finding—the income gradient is robust across specifications, measures, and thresholds—but they bound what we can claim.

8 Conclusion

Can all of Barcelona’s residents walk to essential services within 15 minutes? More or less—but it depends on what you need and where you live. We find a real income–accessibility gap that clusters in space and comes from specific services. Healthcare, schools, and parks concentrate in wealthier central areas; groceries and transit spread evenly. Per capita, the gap vanishes—wealthier areas attract more services *and* more people.

Three takeaways for policy. First, Barcelona should target *specific services*—healthcare and schools above all—in the Low-Low clusters we identify (Figure 4), rather than chasing aggregate accessibility numbers. Second, our market-access result tells planners to think about a neighbourhood’s position relative to the whole city, not just its distance from the centre. Third, the per-capita finding serves as a useful sanity check: more services in rich areas does not automatically mean inadequate provision in poor areas.

On a broader level, our results connect a local urban-planning question to the theoretical machinery of quantitative spatial economics. The spatial multiplier in the SDM is not just an econometric convenience—it captures the same propagation logic that drives general-equilibrium models of trade and migration. If Barcelona’s policymakers take the 15-minute city seriously, they need to think spatially: what happens in one neighbourhood ripples outward to its neighbours.

9 References

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